

A Survey on Unifying Large Language Models and Knowledge Graphs for Biomedicine and Healthcare

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Abstract

In recent years, the landscape of digital biomedicine and healthcare has been reshaped due to the disruptive breakthroughs in AI-facilitated by tremendous data and high-performance computers, large language models (LLMs) have transformed information technology from accessing data to performing analytical tasks. While demonstrating unprecedented capabilities, LLMs have been found unreliable in tasks requiring factual knowledge and rigorous reasoning. Biomedicine and healthcare, as an important vertical domain rapidly benefitting from progress in AI, necessitates strict requirements on the accuracy, controllability, and interpretability of analytical models, posing critical challenges for LLMs. Despite recent studies addressing the hallucination problem of LLMs, research on empowering LLMs with the ability to plan, reason, and ground with explicit knowledge has also started to prosper, especially in the biomedicine and healthcare domain. On the other hand, biomedical data are enormous and notoriously complex, coming from various sources (e.g., biomedical knowledge bases, online literature, and hospitals) and bearing various modalities (e.g., tables, texts, images and time-series). Healthcare professionals have spent decades collecting, cleaning, and curating various types of data. The processes are extremely costly, producing various datasets with different data schemas, coding systems, and quality standards, many privately

owned by the creators, making their integrative analysis and utilization through unified AI techniques still rather challenging. The generalizability of LLMs across different types of data endow them strong promises in automating the processing of large-scale complex healthcare data such as into unified knowledge graphs (KGs). Our goal in this survey is to systematically investigate and summarize recent studies on the unification of LLMs and KGs, towards fully utilizing the value of complex data, unleashing the power of generative AI, and expediting next-generation AI for biomedicine and healthcare applications.

CCS Concepts

• **Applied computing** → **Life and medical sciences**.

Keywords

large language model, knowledge graph, biomedical sciences, health informatics

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1 Introduction

Large language models (LLMs) have reshaped AI research and implementations, with unprecedented capabilities widely shown in various text-related tasks, bringing humans ever close to general AI. Recent research on multi-agent systems have further magnified LLMs' advantages of *language comprehension*, *broad knowledge* and *generalizability* through conversations, showing strong promises for deep human-model collaboration for critical applications [1]. In biomedicine and healthcare, extensive enthusiasm

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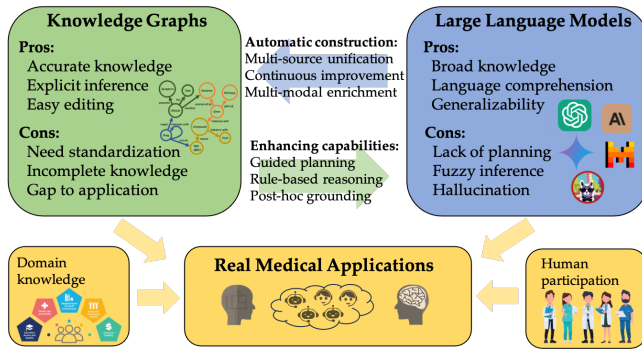


Figure 1: Overview of the proposed biomedical knowledge language models framework.

has been witnessed on the exploration and evaluation of LLMs in answering medical questions [2], extracting clinical information [3] and assisting clinical decisions [4]. Studies have also revealed the limitations of LLMs regarding their *lack of knowledge* [5], *fuzzy inference* [6], and *hallucination* [7]. Specifically, in biomedicine and healthcare, the lack of knowledge can be caused by the lack of access to high-quality data about various biomedical concepts and patient conditions, as well as the rapidly evolving new biomedical knowledge; the fuzzy inference nature can lead to difficulties in conducting reliable comprehension and stable predictions for complex medical questions; and hallucination creating factual errors and misinformation can cause fatal and life-threatening problems in the healthcare workflows [8].

Knowledge graph (KG) has been widely studied across academia and industry, due to its advantages in storing *accurate*, *explicit* and *easily-modifiable* knowledge [9]. In biomedicine and healthcare, researchers and professionals have spent decades collecting, processing, and curating various types of biomedical and clinical data towards the construction of medical KGs [10], which are widely used to support basic science research [11], pharmaceutical research [12], clinical decisions [13] and policy making [14]. However, biomedical data are notoriously noisy and complex, where datasets about specific concepts and conditions come from various sources such as institutions using different data schemas and coding conventions [15], and the data can also include multiple modalities such as tables, texts, images, and time-series [16]. While such *multi-source* and *multi-modality* data hold great promises in integrative and comprehensive biomedical analysis, extracting and unifying high-quality knowledge from them is non-trivial.

Recently, significant research attention has been drawn to the synergies between KGs and LLMs [17], due to their naturally complementary advantages (Figure 1). The construction and modeling of KGs have always relied on advances in natural language processing (NLP) tools, and nowadays, researchers have intensively explored language models towards the embedding [18], completion [19] and construction [20] of KGs. Studies in the recent years have also bloomed to explore the utilization of KGs for enhancing LLMs through providing new sources of knowledge during pre-training [21] or inference [22], and enabling knowledge-based interpretation and evaluation [23]. In very recent years, pioneering studies have also started to explore the combination of KGs and LLMs for biomedicine and healthcare [24]. Most of these studies

have focused on specific healthcare applications and only implemented shallow and straightforward technical designs, without fundamentally improving the KGs and LLMs.

To expedite LLM-based research towards next-generation AI for health, this survey will comprehensively investigate and summarize recent works addressing the *data*, *model* and *application* challenges of unifying KGs and LLMs for biomedicine and healthcare, through a systematic conceptual framework of BioMedKLM (BioMedical Knowledge Language Models, Figure 1). We present BioMedKLM to discuss major functionality needed to build high-quality KGs that integrate complex *biomedical data*, enhance LLMs to obtain reliable *biomedical models*, and properly employ the data and model to enable critical and novel *biomedical applications*. This survey, as illustrated in Figure 1, consists of three distinct but interrelated research perspectives that address the data, model and application challenges blocking the transformation of AI for health.

- **Perspective P1: LLM-aided KG construction from multi-source multi-modality biomedical data.** We will discuss LLM-based methods for unifying existing biomedical KGs collectable from different sources (P1.1), continuously enhance biomedical KGs by extracting concepts and relations from evolving biomedical literature, and (P1.3) comprehensively augment biomedical KGs through the integration of multi-modal biomedical data.
- **Perspective P2: KG-guided LLM enhancement towards reliable biomedical models.** We will discuss the utilization of biomedical KGs to enhance the capabilities of LLMs through providing biomedical knowledge to enhance LLM planning (P2.1), enabling biomedical neural symbolic reasoning with LLMs (P2.2), and enforcing post-hoc biomedical error detection to verify LLMs’ knowledge and reasoning (P2.3).
- **Perspective P3: Knowledge language co-modeling to facilitate reliable biomedical applications.** We will discuss how the integration of KGs and LLMs enables critical biomedical applications including clinical decision support for enhanced diagnostics (P3.1), drug discovery and development for accelerated pharmaceutical research (P3.2), and biomedical knowledge management for organizing the vast body of biomedical information (P3.3).

This survey establishes a systematic and comprehensive framework of BioMedKLM that aims to enable the synergistic and progressive improvements of KGs and LLMs for biomedicine and healthcare. Techniques related to BioMedKLM reflect methodological innovations in data mining and generative AI, fundamentally improving KGs via automatic extraction and integration of multi-source multi-modality knowledge, enhancing LLMs towards knowledge-guided planning, reasoning and grounding, and unleashing the power of generative AI towards ethical, trustworthy and human-centered biomedical applications.

2 LLM-aided KG Construction

In this section, we examine the benefits of leveraging LLMs for medical KG construction, particularly in *accuracy*, *consistency*, *coverage*, and *freshness* of knowledge. Traditional approaches, as surveyed in [10], often focus on specific disease areas or entity types. Efforts to

build comprehensive biomedical KGs primarily rely on integrating existing sources [25], using coding systems for entity alignment [26]. However, these methods struggle with terminological variations, leading to redundancy and inconsistency. A few studies have attempted to construct general-purpose medical KGs through integrating existing ones [25, 27], but they heavily rely on existing coding systems and thesauruses [26] for entity alignment across KGs, which often fail in front of varying terminologies such as due to different conventions or abbreviations, leading to high degrees of duplication and inconsistency. Recently, several studies started to explore the potential of LLMs to automate KG construction [20, 28]. In the following, we discuss key applications of LLMs and multi-modal foundation models (MMFMs) toward constructing high-quality KGs using multi-modality biomedical data from existing KGs, biomedical literature and medical institutions.

2.1 Integrating existing KGs

KG integration, often referred to as knowledge fusion or alignment, involves merging KGs from diverse sources and formats. It is a core challenge in knowledge engineering, requiring effective strategies to ensure consistency and interoperability [29]. KG integration, also known as knowledge fusion or knowledge alignment, represents a fundamental challenge in the broader landscape of knowledge engineering, which involves integrating multiple KGs that originate from varied sources and formats [29]. While individual KGs often excel in specific domains or use cases, their true potential can be unlocked through effective integration, enabling more comprehensive and robust knowledge representation [27]. As the number and diversity of KGs continue to grow, the need for effective integration methods becomes increasingly critical. However, the integration of existing KGs faces several key challenges: (1) *semantic heterogeneity across sources*: Different KGs often use varying terminologies, definitions, and contextual frameworks to represent similar concepts; (2) *varying granularity levels in knowledge representation*: KGs may differ in the detail and depth with which they describe entities and relationships, impacting the consistency and usability of integrated data. Although neural approaches have been proposed for entity alignment on KGs, these methods generally depend heavily on labeled data for training. However, obtaining sufficient labeled data often involves substantial manual effort and can be rather costly. LLMs have emerged as a promising solution to these challenges with unique advantages: First, their strong natural language understanding capabilities enable them to capture semantic relationships among concepts that may be missed by traditional string-matching or embedding-based approaches. Second, LLMs can draw on their extensive knowledge acquired during pre-training to aid in disambiguating entities and mapping relationships across different KGs. Third, LLMs possess robust few-shot learning abilities, making them particularly valuable for specialized domain applications where labeled data are limited.

Prior works [30, 31] explore the potential of linking biomedical entities across KGs. Specifically, HiPrompt [30] aligns entities between biomedical KGs and standardized ontologies via a two-stage approach: traditional information retrieval techniques (BM25) followed by a LLM-based re-ranking using hierarchy-oriented prompts. PromptLink [31] further improves this two-stage framework by

first eliciting the biomedical prior knowledge from the LLM for the concept linking task and then enforcing the LLM to reflect on its own predictions to further enhance their reliability. The improvements are especially significant for weaker LLMs, which is intuitive and useful since not every medical institution can always (safely) access the strongest LLMs. Besides, AutoAlign [32] utilizes off-the-shelf LLMs to capture relationships between entity types with a predicate-proximity graph and then aligns entities across KGs by computing similarity in the embeddings space.

The above advances in LLM-aided KG integration suggest several promising future directions. For example, future LLM-driven KG integration could focus on enabling *evolving knowledge graph updates* by resolving inconsistencies between new and existing knowledge. Another key challenge in leveraging LLMs to enrich KGs is the risk for misinformation [33]. To alleviate this, human-in-the-loop frameworks play a critical role in verifying and refining LLM-generated outputs [34].

2.2 Constructing and Completing KGs

KGs have high-standard requirements on the quality of knowledge, regarding accuracy, consistency, coverage and freshness. No matter constructed through manual curation, NLP tools, or their combinations, KGs can unavoidably include erroneous knowledge. Moreover, when multiple KGs are integrated, conflicting knowledge can emerge. Finally, new knowledge is constantly generated from new experiments and research, making existing knowledge inaccurate and incomplete. LLMs have emerged as a promising solution, leveraging the vast and adaptable knowledge acquired during pre-training to overcome these limitations.

The key advantage of LLMs in this domain lies in their ability to generate novel, semantically coherent information that can supplement and enrich existing KGs. Unlike rule-based or supervised machine learning approaches, LLMs can leverage their extensive understanding of language and the world to infer missing connections, identify new entities, and uncover implicit relationships - all without being constrained by the limitations of manually curated training data.

As demonstrated in recent works, LLM-based approaches have shown strong potential in KG construction and completion. Zhu et al. [20] exploit in-context learning to predict missing entities and relations, generating new triplets to expand existing KGs. Meanwhile, KC-GenRE [35] frames KG completion as a retrieval and ranking problem. Their approaches first retrieve candidate entities and then employ LLMs to refine and reorder them, ultimately forming additional knowledge triplets. Zhang et al. [36] further incorporate the schema elements relevant to the prompt for standardizing the triplets and improving the generation quality of LLMs. Inspired by recent progress of LLM reasoning, Nie et al. [37] further chain-of-thought promoting techniques [38] to better guiding LLMs in understanding triple knowledge in unstructured data with improved triplet extraction accuracy.

Beyond traditional text-based prompting, code-based instructions offer an alternative approach for guiding LLMs in structured knowledge generation. Code LLMs, designed for processing structured data like programming code, naturally align with the hierarchical and relational nature of KGs. Their training on structured

inputs enables them to better capture and manipulate graph-based representations, making them well-suited for tasks requiring logical consistency and precision [39, 40]. For ontology expansion task, CodeTaxo [41] represents entity relationships with a hierarchical structure inspired by programming syntax for better harnessing LLMs' ability to interpret structured patterns. By leveraging code-like representations, it enables LLMs to systematically construct taxonomies, improving the organization and completeness of KGs. CodeKGC [42] encoded the schema of KGs by modeling code definitions for capture the structural information inherent in the data. By leveraging chain-of-thought prompting, it systematically generates precise knowledge triples. To summarize, code-driven prompting enables LLMs to structure and categorize concepts more efficiently by leveraging syntax-based reasoning, improving KG organization and consistency. This method outperforms conventional natural language prompts by providing a more structured and logic-driven way to enhance KG completion.

Apart from prompting, several studies explored fine-tuning to adapt LLMs for KG completion. KG-LLM [43] directly performs instruction tuning on KG completion tasks including *triplet classification*, *relation prediction* and *link prediction* and outperform frontier models using lightweight backbones only. KOPA [44] first conducts pretraining to obtain entity and relation embeddings, which transforms them into virtual knowledge tokens within a unified textual space. These tokens then act as prefixes in LLM prompts that enable structure-aware reasoning by combining LLM generation with KG-based retrieval. MKGL [45] structures knowledge as three-word sentences (entity-relation-entity triplets), and then finetunes LLMs to generate and complete KG triplets by leveraging real-time KG context retrieval and token embedding augmentation to enhance factual consistency. KG-FIT [46] attempts to exploit open-world knowledge from LLMs to enhance KG embeddings. It first constructs a semantically coherent, hierarchical structure of entity clusters, then fine-tunes these embeddings by integrating the hierarchical structure within textual embeddings. Such a hybrid approach combines structural KG information with semantic depth from LLMs for richer representations.

2.3 Enriching KGs with multi-modality data

Another property of biomedical KGs is the diverse sources and modalities of useful knowledge. Besides existing KGs and online literature, medical institutions including hospitals, clinics and medical centers generate vast amounts of patient data daily, through patient visits, clinical trials, health monitoring devices, and so on. Traditionally, specialized models and algorithms have been developed to process and analyze various modalities of patient data such as electronic health records (EHRs), medical images, physiological waveforms, clinical notes and health insurance claims. These methods can hardly perform integrative analysis across data modalities and generalize across different medical institutions. Recently, LLM-based multimodality foundation models (MMFMs) have shown strong promises in analyzing multi-modality data through the unified interface of languages [47]. However, studies on aligning general MMFMs to real patient data have shown this task to be rather challenging due to the lack of high-quality fine-grained pairs of *X-and-text* labeled data for instruction tuning—let *X* be *chest x-rays*

and *text* be *radiology reports*, most reports do not include accurate annotations (bounding boxes) of lesions in the lung [48]; let *X* be *spatiotemporal EEG/SEEG signals* and *text* be *nursing notes*, most notes do not indicate accurate time ranges and spatial locations of epilepsies in the brain [49]. Besides the development of biomedical MMFMs, how to utilize them for the extraction of novel knowledge from multi-modality patient data that can be properly incorporated into the biomedical KGs for further integrative utilization towards empowering LLMs and facilitating various downstream biomedical applications can highly impact AI practices in health informatics, but this remains much under-explored.

To adapt MMFMs with domain-specific knowledges, BioMed-VITAL [48] is designed for efficiently aligning biomedical MMFMs with clinician preferences. Through novel designs in the three steps of *data generation*, *selection* and *instruction tuning*, BioMed-VITAL demonstrates significant improvements in medical visual chat and VQA datasets, showing strong promises for further utilization towards open-ended knowledge discovery from medical images. On the other hand, OpenVik [50] explores prompting MMFMs to generate format-free visual knowledge from the detected regions. OpenVik further gathered novel relational knowledge generated from images from Visual Genome [51] to form a novel KG, and demonstrated its advantages over existing commonsense KGs such as ConceptNet [52] and COMET [53] for various KG-supported downstream tasks. It opens up the new arena of utilizing MMFMs for automatic relational knowledge discovery. PromptCap [54] leverages pre-trained multimodal large language models (MLLMs) to understand both visual and textual information in a unified framework for improved visual relation extraction. Yang et al. [55] introduce an automated pipeline for building product KGs in e-commerce directly from raw images. It leverages MMFMs to extract visual details and utilizes an LLM to infer missing attributes and relationships. By hierarchically structuring and linking entities, this pipeline enables scalable KG construction without requiring manual intervention.

3 KG-guided LLM Enhancement

LLMs have shown impressive communication and question answering capabilities, demonstrating strong promises in various healthcare applications [2, 56]. However, to reliably model biomedical data and generate factual and accurate answers, LLMs still face the challenges of lacking domain knowledge, fuzzy inferences, and hallucination [8]. Retrieval augmented generation (RAG) [57], which aims at retrieving question-relevant evidences and generating evidence-based answers, has strong promises in evidence-critical domains including biomedicine and healthcare. However, effective and efficient RAG for biomedicine and healthcare is seldom studied and requires solutions to practical challenges including (1) how to find relevant evidences from complex biomedical data, (2) how to conduct biomedically valid complex inference, and (3) how to reliably guarantee the removal of biomedical errors. In this section, we will comprehensively investigate these challenges and demonstrate the advantages of utilizing KGs to enable reliable LLMs for biomedicine and healthcare.

3.1 Planning with domain knowledge

Modeling biomedical data is challenging because diseases are often complicated and heterogeneous such as regarding the causes, symptoms, and treatment effects. Given a question seemingly as simple as ‘Will patient 0315 with type 2 diabetes develop cardiovascular comorbidities in the next 5 years’, an experienced doctor will naturally expand the question into a set of related questions– e.g., ‘Does the patient suffer from obesity’, ‘Does the patient take metformin’, ‘Does the patient smoke’, ‘What is the patient’s current and historical blood pressure’, etc., so as to find and retrieve necessary evidences for answering the original question. This process is referred to as query planning in LLM research [58], and LLMs are known to perform poorly and hallucinate when planning for complex queries even with explicit and deliberately designed prompts [59]. Moreover, LLMs also lack concrete knowledge about biomedicine and healthcare to generate faithful retrieval plans– e.g., for calling tools to retrieve existing patient data, asking the patient through a chatting interface, or ordering new tests for the patient, especially facing the complexity of diseases and biomedical data.

Early works have primarily relied on external retrievers to acquire relevant factual knowledge to enhance LLM reasoning. For instance, Baek et al. [60] proposed a direct retrieval method to extract pertinent triples from KGs. However, the retriever, being a shallow embedding model, may not consistently retrieve the most relevant facts, particularly in the complex medical domain. KGs encompass a wealth of domain-specific knowledge, posing challenges for LLMs with limited domain expertise in comprehending and utilizing this information for answering medical questions. To further harness the potential of LLMs in leveraging domain knowledge, the *plan-and-solve* paradigm [61] has been introduced, where LLMs are first prompted to generate a plan. Based on the generated plan, LLMs can retrieve the relevant domain knowledge and perform reasoning to generate answers [62]. However, existing approaches fall short in handling the complex structured knowledge within KGs to enable effective planning and reasoning. To address this limitation, Luo et al. [63] proposed a *planning-retrieval-reasoning* framework, RoG, which empowers LLMs to plan and reason over KGs. The overall framework is depicted in Figure 2.

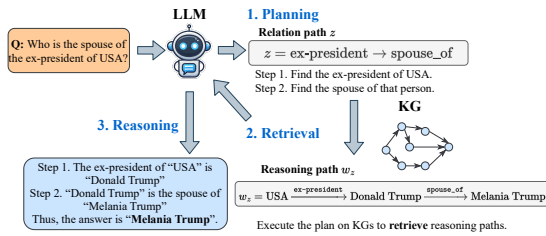


Figure 2: The overall framework of planning and reasoning on KGs (RoG).

RoG first generates multiple relation paths, grounded in knowledge graphs (KGs), which are used as structured plans. Relation paths, which represent semantic relationships between entities, have been widely applied in various reasoning tasks on KGs [64] by decomposing a complex reasoning into multiple simple steps. Leveraging these relation paths, one can efficiently retrieve up-to-date knowledge from KGs through a constrained breadth-first

search. Consequently, relation paths act as reliable plans that guide both the retrieval and reasoning processes on domain-specific KGs. Furthermore, by treating relation paths as plans, one ensures that these plans are firmly grounded in KGs, enabling large language models (LLMs) to retrieve pertinent knowledge and perform accurate reasoning. To formalize this approach, RoG is framed as an optimization problem aimed at maximizing the probability of deriving an answer from a KG $\mathcal{G}$ with respect to a given question q , by generating relation paths z as the guiding plan:

$$P_{\theta}(a|q, \mathcal{G}) = \sum_{z \in \mathcal{Z}} P_{\theta}(a|q, z, \mathcal{G}) P_{\theta}(z|q), \quad (1)$$

where θ denotes the parameters of LLMs and a denotes the final answer. To enable accurate planning with domain knowledge, two instruction tuning tasks are designed: 1) *planning optimization*, which distills the knowledge from KGs into LLMs to generate faithful relation paths as plans; 2) *retrieval-reasoning optimization*, which enables LLMs to reason based on the retrieved reasoning paths. The final objective function of RoG is the combination of the planning optimization and retrieval-reasoning optimization, which can be formulated as

$$\mathcal{L} = \underbrace{\log P_{\theta}(a|q, \mathcal{Z}_K^*, \mathcal{G})}_{\text{Retrieval-reasoning}} + \underbrace{\frac{1}{|\mathcal{Z}^*|} \sum_{z \in \mathcal{Z}^*} \log P_{\theta}(z|q)}_{\text{Planning}}, \quad (2)$$

where the shortest paths $\mathcal{Z}^* \subseteq \mathcal{Z}$ between q and a in KGs are used as supervision signals. The probability of LLMs generating faithful relation paths is maximized through distilling the knowledge from KGs. In this way, with the proposed RoG, LLMs can effectively retrieve domain knowledge from KGs with planning, which significantly enhances the reasoning capability of LLMs.

3.2 Reasoning with structured knowledge

After retrieving evidences, LLMs need to follow them and generate answers. While evidence following has been shown achievable through prompt designs [65, 66], LLMs will struggle to generate the correct answers when important supporting evidences are missing, which is very common in biomedicine and healthcare due to the sparse and incomplete patient data especially regarding rare diseases. With strong prior knowledge of LLMs and available partial evidences, it is promising to infer missing evidences with LLMs between the retrieval and generation steps, which ideally requires the LLMs to be able to conduct accurate and efficient biomedically valid reasoning. Recently, extensive research has been conducted on enabling LLMs for knowledge-based reasoning, which can be roughly divided into embedding-based (neural) approaches [67] and rule-based (symbolic) approaches [68]. However, the neural approaches loose interpretability and can hardly work with sparse KGs, while the symbolic approaches are resource-consuming due to large amounts of paths that do not encode valid logic rules. LLMs are neural models that can also naturally work with embeddings, making it an ideal backbone for combining neural and symbolic approaches towards accurate and efficient reasoning, which, however, is hardly studied.

Recent efforts have focused on enabling LLMs to perform reasoning on structured KGs through retrieval-based methods and prompting techniques [68]. CoK [69] and KD-CoT [70] retrieve

facts from external KGs to guide the chain-of-thought (CoT) reasoning process conducted by LLMs. To capture graph structures, GNN-RAG [71] utilizes a lightweight graph neural network to efficiently retrieve knowledge from KGs, formatted as sentence paths to stimulate the reasoning process in LLMs. Mindmap [72] introduces a prompt-based approach that equips LLMs with the ability to comprehend and reason over KGs. Despite the success of these methods, challenges persist in the development of principled prompts for KG representation and reasoning. Furthermore, LLMs continue to face limitations in their understanding of graph structures and reasoning with text-based graph prompts [73].

Unlike previous approaches that necessitate a computationally intensive fine-tuning phase or the design of ad-hoc prompts for LLMs, Luo et al. [74] recently proposed the KG-constrained reasoning (GCR) paradigm. GCR integrates unstructured reasoning in LLMs with structured knowledge in knowledge graphs (KGs), aiming to enable efficient and effective reasoning over structured knowledge. The overall framework is depicted in Figure 3.

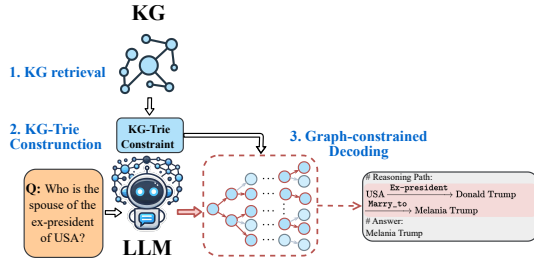


Figure 3: The overall framework of KG-constrained reasoning (GCR).

Graph-constrained reasoning, inspired by the concept that LLMs reason through decoding [38], incorporates the KG structure into the LLM decoding process. This enables LLMs to directly reason on graphs by generating reliable reasoning paths grounded in KGs that lead to correct answers. Specifically, given a question, a retrieval module is first adopted to find a relevant KG that is helpful for reasoning. Then, the KG is converted into a structured index, KG-Trie, to facilitate efficient reasoning on KG using LLMs. Trie is also known as the prefix tree [75] that compresses a set of strings, which can be used to restrict LLM output tokens to those starting with valid prefixes. KG-Trie encodes the reasoning paths in KGs as formatted strings to constrain the decoding process of LLMs. Then, graph-constrained decoding is proposed that employs a lightweight KG-specialized LLM to generate multiple KG-grounded reasoning paths and answers. With the constraints from KG-Trie, one ensures faithful reasoning while leveraging the strong reasoning capabilities of LLMs to efficiently explore paths on KGs in constant time. In this way, GCR bridges the gap between structured knowledge in KGs and unstructured reasoning in LLMs, allowing for efficient reasoning on KGs via LLM decoding.

3.3 Reflecting with atomic knowledge

Biomedicine and healthcare have high requirements on factuality and accuracy. While the previous subtasks can improve factuality via planning and accuracy via reasoning, errors can still occur due to LLM’s failure to follow evidences, misunderstanding and

hallucination. Therefore, we propose to utilize KGs to add post-hoc knowledge grounding, further ensuring the LLM reliability. While existing works on KG-based RAG mostly use KGs as additional resources of factual knowledge [76], post-hoc error detection will be conducted on the LLM outputs. This is fundamentally different and more challenging, as the outputs often blend facts from diverse sources and involve multiple reasoning steps.

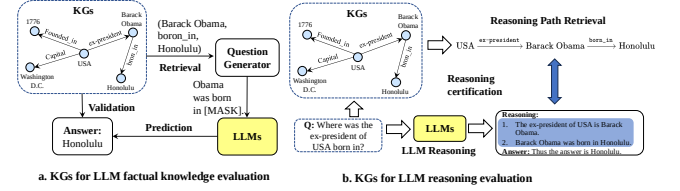


Figure 4: The illustration of LLM reflection with KGs. (a) The evaluation of the factual knowledge inside LLMs. (b) The evaluation of the reasoning process of LLMs with KGs.

The hallucination phenomenon in LLMs is commonly attributed to their limited factual knowledge. To systematically evaluate the factual knowledge embedded in LLMs and enable faithful respond in solving medical problem, we propose a novel framework, as illustrated in Figure 4a, which leverages Knowledge Graphs (KGs) [77]. In contrast to traditional approaches that rely on human-annotated question-answer datasets, our method generates valid and diverse questions from KGs at varying levels of difficulty while ensuring comprehensive knowledge coverage. Specifically, we extract atomic knowledge from KGs in the form of sets of triples. These triples are then converted into question-answer pairs using various question generation techniques, such as template-based and LLM-based methods. The generated question-answer pairs are subsequently used to assess the factual knowledge of LLMs by comparing the model-generated answers with the corresponding ground-truth answers. The evaluation outcomes provide insights into the factual accuracy of LLMs and can be used to better understand their hallucination behaviors. This approach facilitates a systematic evaluation of LLMs’ factual knowledge and offers valuable guidance for enhancing their reliability across a range of high-stakes applications.

In addition to factual knowledge, the structure of knowledge graphs (KGs) can be leveraged to justify the reasoning process of large language models (LLMs). Minh-Vuong et al. [78] developed a framework that explores the chain-of-thought (CoT) reasoning capabilities of LLMs in multi-hop question answering by utilizing KGs, as illustrated in Figure 4b. The framework includes two evaluation modules: discriminative evaluation and generative evaluation. The discriminative evaluation assesses whether LLMs possess sufficient knowledge to conduct faithful reasoning. It inputs both valid and invalid reasoning paths, retrieved from KGs, into LLMs and requests them to predict the validity of these paths. Conversely, the generative evaluation aims to assess the faithfulness of the LLMs’ reasoning process by grounding it in KGs. For a reasoning process generated by LLMs, the generative evaluation module retrieves relevant facts from KGs and compares them with the ground-truth reasoning paths. The evaluation results serve to reflect the reasoning capabilities of LLMs and offer insights into the faithfulness of

their reasoning. Despite demonstrating impressive reasoning abilities, LLMs face ongoing challenges in ensuring faithful reasoning, particularly in multi-hop question answering.

4 Applications in Biomedicine and Healthcare

Biomedicine and healthcare represent ideal domains for the integration of KGs and LLMs due to their critical need for both structured knowledge and natural language understanding. The motivation for applying MedKLM in these domains stems from the complexity of biomedical data, the high stakes of healthcare decisions, and the necessity for interpretable AI systems that clinicians can trust. Healthcare applications face unique challenges including data fragmentation across institutions, strict requirements for accuracy and reliability, privacy concerns, and the need to incorporate domain-specific expertise. Despite these challenges, BioMedKLM offers promising directions for transforming healthcare through: (1) enhancing clinical decision support with reasoning grounded in biomedical knowledge; (2) accelerating drug discovery and development by connecting molecular structures with biomedical insights; and (3) improving biomedical knowledge management to organize, integrate, and apply the vast amount of healthcare information available. The following subsections explore these specific applications where the synergy between KGs and LLMs has shown particular promise in addressing complex healthcare needs.

4.1 BioMedKLM for clinical decision support

Clinical decision support represents a key application domain for BioMedKLM, where the synergy between KGs and LLMs enables more accurate and interpretable clinical decisions. By combining structured biomedical knowledge with advanced language understanding capabilities, these systems help clinicians analyze complex patient data and make informed medical decisions. We categorize existing approaches into two categories:

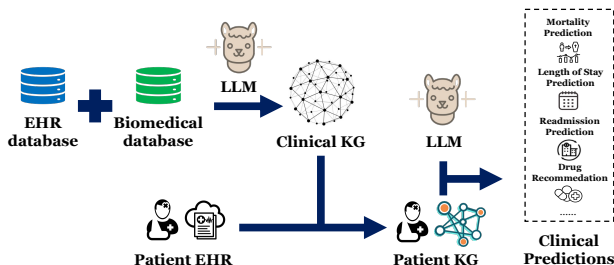


Figure 5: An example application of BioMedKLM for clinical decision support.

(1) Integrated KG+LLM approaches. Recent developments demonstrate how the integration of KGs and LLMs can enhance clinical tasks through complementary strengths. GraphCare [79] uses LLMs to construct patient-specific KGs, enabling more personalized and accurate clinical predictions. RAM-EHR [80] integrates a hyper-graph that captures complex biomedical concept relationships and LLM-summarized medical context to improve clinical predictions. EMERGE [81] demonstrates effective fusion of KGs with clinical notes for enhanced prediction accuracy. MedIKAL [24] reimagines the biomedical KG as an assistant that guides LLM decision-making, whereas KARE [82] introduces a novel framework combining LLM’s

reasoning capability and precise biomedical knowledge retrieval with its constructed comprehensive biomedical KG community summaries. ComLLM [83] introduces a method prompting LLMs with a disease-specific KG for disease progression prediction, while MedTok [84] introduces multi-modal (KG and LLM) medical code learning to enhance clinical predictions. MedRAG [85] combines a four-tier hierarchical diagnostic KG with RAG to enable accurate diagnosis of diseases with similar manifestations, while also providing personalized treatment recommendations and proactive follow-up questions. We show an example of using BioMedKLM for clinical predictions in Figure 5.

(2) KG/Graph-only and LLM-only foundations. Earlier approaches focused on either structured knowledge or language understanding independently. Graph-based methods [86] established the fundamental importance of structured biomedical knowledge representation for clinical predictions. These methods could be significantly enhanced by incorporating LLMs for automated graph construction and natural language explanation generation. Conversely, pure LLM approaches [87, 88] and domain-specific language models [3] demonstrated remarkable natural language understanding capabilities in the medical domain, but their potential could be further realized through integration with KGs for improved factual accuracy and reasoning capabilities.

4.2 BioMedKLM for drug discovery and development

Drug discovery and development represents another key application domain for BioMedKLM, where the synergy between KGs and LLMs enables more accurate and efficient drug design and evaluation.

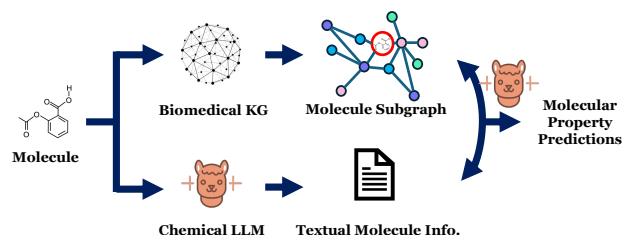


Figure 6: An example application of BioMedKLM on molecular property prediction tasks for drug discovery.

(1) Drug discovery. Molecules inherently possess graph structures, making them natural candidates for graph-based representation learning. Recent work has explored various approaches to enhance molecular representations [89], with a particularly promising direction being the integration of knowledge graphs [90]. For instance, Ye et al. [91] pioneered the incorporation of static KG embeddings into molecular fingerprints to improve drug-target interaction prediction. KANO [92] constructed Element-KG that augments molecular representations at the chemical element level through contrastive learning. Taking a different approach, Gode [90] developed a comprehensive biomedical KG that treats molecules as graph-in-graph structures, enabling deeper integration of biomedical knowledge into molecular representations. The field has recently seen exciting developments in leveraging LLMs’ reasoning capabilities to

enhance molecular property predictions [93], opening new opportunities for BioMedKLM to synergistically combine KGs and LLMs in drug discovery applications. Figure 6 illustrates an example of applying BioMedKLM to drug discovery, where we generate molecular representation by integrating structured knowledge from biomedical KGs [90] with textual understanding from chemical LLMs [94]. This knowledge-enriched representation is then learned by LLM to make accurate and interpretable property prediction for highly reliable drug discovery.

(2) Drug development. Drug development requires deep understanding of biological mechanisms, disease pathways, and drug effects. Recent advances leverage both KGs and language models to accelerate this process. KGWAS [95] and TxGNN [96] exemplified this trend as graph foundation models: KGWAS used functional genomics KGs to detect disease-associated genetic variants in rare diseases, while TxGNN learned from biomedical KGs for drug repurposing. These systems could be enhanced by using LLMs to enrich their knowledge graphs with literature-derived relationships and translate graph reasoning into natural language explanations. Malas et al. [97] demonstrated KG-based drug prioritization by extracting semantic features between drugs and diseases to train a classifier that successfully identified promising candidates for polycystic kidney disease. In clinical trials, LINT [98] combined medical code KGs with LLM embeddings to predict trial outcomes. While TrialGPT [99] demonstrated strong performance using LLMs alone for trial matching, its retrieval-matching-ranking pipeline presents promising opportunities for incorporating KGs to further enhance matching accuracy and interpretability in drug evaluation studies.

4.3 BioMedKLM for biomedical knowledge management

Biomedical knowledge management is a critical aspect of modern healthcare, requiring systematic organization, curation, and integration of information from diverse sources such as clinical guidelines, research literature, and patient data. The combination of LLMs and KGs offers a powerful framework for automating knowledge extraction, integration, and maintenance while ensuring accuracy, consistency, and interpretability. This section explores the application of BioMedKLM for biomedical knowledge management.

Biomedical Knowledge Graph Construction and Updates. Construction and maintenance of biomedical KGs are fundamental to many healthcare applications. BioMedKLM facilitates automatic construction and continuous updating of KGs through several approaches: Graphusion [100] utilizes LLMs for KG construction with a global perspective, employing retrieval-augmented generation (RAG) to extract and integrate knowledge from diverse sources. KG-RAG [101] optimizes prompt generation for LLMs to enhance accuracy and relevance of extracted biomedical knowledge. Domain-specific KGs have been developed using LLMs, such as KGs for diabetes [102], traditional Chinese medicine [103], heart failure [104], rare disease [105], and Alzheimer’s disease research [106].

Data Generation and Knowledge Application. BioMedKLM extends beyond knowledge organization to practical applications: (1) ClinGen [107] leverages LLMs to generate synthetic clinical text data while ensuring accuracy and alignment with real-world scenarios. (2) Clinical decision support systems integrate KGs and

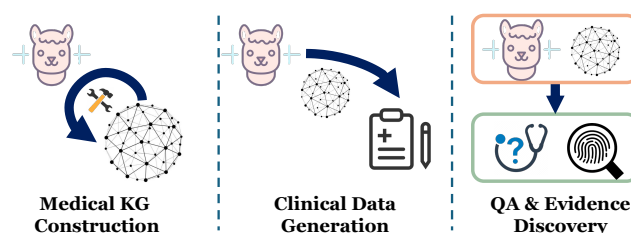


Figure 7: An example application of BioMedKLM for biomedical knowledge management.

LLMs to enhance prediction accuracy and interpretability [82, 108]. (3) Risk prediction models utilizing KGs have been developed for various conditions such as macular edema [109], adverse drug reactions [110], and cancer [111].

Question Answering and Evidence Discovery. BioMedKLM plays a crucial role in medical question answering and knowledge validation: (1) Studies [112, 113] demonstrate effective QA systems using LLM-KG combinations, including specialized systems for hepatitis B [114] and hypertension [115]. KGAREview [116] combines LLMs with KGs for medical QA through an iterative generate-verify-refine pipeline, improving accuracy over existing methods by dynamically refining generated answers using structured knowledge from KGs. This integration enables extraction of complex relationships and evidence from biomedical literature. (2) The framework shows potential in detecting medical fraud and waste, analyzing patterns in healthcare data to identify potential cases of fraud [117, 118]. BioMedKLM also leverages LLMs and KGs to automate the process of literature synthesis [119], enabling researchers and clinicians to efficiently identify and integrate key findings from vast amounts of text.

This integration of LLMs and KGs in biomedical knowledge management demonstrates significant potential for improving healthcare information organization, accessibility, and application, while maintaining high standards of accuracy and reliability.

5 Conclusions and Future Directions

In this paper, we discuss the trending efforts of knowledge and language co-modeling. We showcase promising attempts to utilize LLMs to automate the construction, integration, and enrichment of KGs, and discuss how KGs can help with planning paths, guide reasoning with structure, and ground knowledge with reflection, enhancing the reliability of LLMs. We also provide a detailed review of knowledge and language co-modeling for real-world applications in biomedicine and healthcare. While this nascent research direction holds great potential, we envision several specific problems worthy of further exploration including the development of more powerful and generic KGs, the trade-off between unified and specialized KGs, the evaluation of knowledge in LLMs, and the resolution of conflicts between KGs and LLMs for biomedicine and healthcare.

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